

AP Calc Lesson 2.3 Homework

Name _____

1.
$$f(x) = \begin{cases} g(x), & x \leq 4 \\ h(x), & x > 4 \end{cases}$$

Let f be a function defined as shown above.

a. What must be true in order to justify that f is continuous at $x = 4$?

b. What must be true in order to justify that f is differentiable at $x = 4$?

-  2. The number of attendees seated at a school play is modeled by a differentiable function A , where t is measured in minutes after the pre-show call is made. The function A is given by $A(t) = 2000(1 - e^{-0.8t})$. Find the rate at which attendees are being seated, in attendees per minute, 3 minutes after the pre-show call has been made.

3.
$$f(x) = \begin{cases} h(x), & x \leq 1 \\ 3x + 4, & x > 1 \end{cases}$$

Let function f be the piecewise function defined above.

a. Let $h(x) = 7$. Determine whether f is differentiable at $x = 1$. Explain your reasoning.

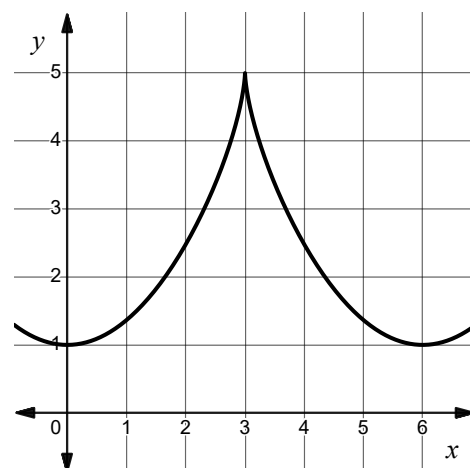
b. Let $h(x) = 3x - 5$. Determine whether f is differentiable at $x = 1$. Explain your reasoning.

c. Let $h(x) = x^2 + x + 5$ and h' , the derivative of h , be given by $h'(x) = 2x + 1$. Determine whether f is differentiable at $x = 1$. Justify your answer.

4. Let function f be defined by $f(x) = 5 - 4 \left(\cos \left(\frac{\pi x}{6} \right) \right)^{2/3}$. The graph of $y = f(x)$ is shown.

a. Use technology to find $f'(2)$, $f'(3)$, and $f'(4)$ or state that the value does not exist.

b. How does the graph of f support your answers to part a?

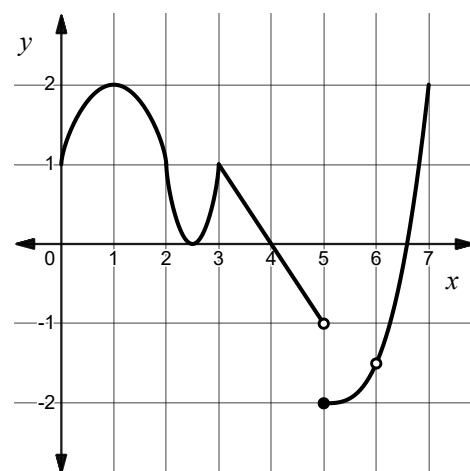


5. The figure shows the graph of function $y = f(x)$, which has a vertical tangent at $x = 2$ and horizontal tangents at $x = 1$ and $x = 2.5$.

a. Identify the x -coordinate of each point on the graph of f such that f is not differentiable. Give a reason for each answer.

b. Determine the slope of the line tangent to f at $x = 4$.

c. Evaluate $\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1}$.



6.

x	$f(x)$	$f'(x)$
1	-3	2
2	1	4
4	9	1

Selected values of a continuous function f and its derivative f' are given in the table.

Function h is given by the piecewise function defined by $h(x) = \begin{cases} 4x - 7, & x \leq 2 \\ f(x), & x > 2 \end{cases}$. Determine whether function h is differentiable at $x = 2$. Justify your answer.



7.

Let f be a differentiable function given by $f(x) = \begin{cases} 5 \tan\left(\frac{x}{4}\right) + 2, & x \leq \pi \\ cx + b, & x > \pi \end{cases}$ where c and d are nonzero constants.

a. Find $f(\pi)$.

b. Use technology to find $\lim_{x \rightarrow \pi^-} f'(x)$.

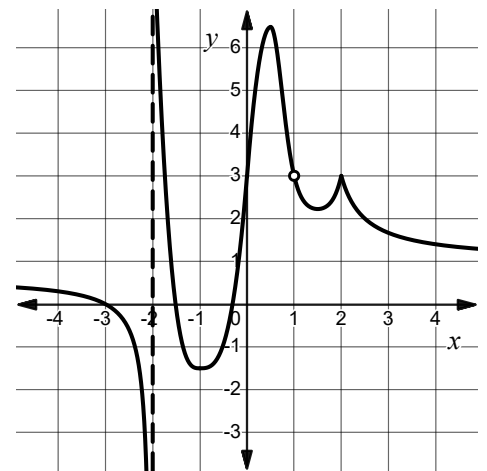
c. Use your answers to parts a and b to find the values of c and d .

8. The graph of f' , the derivative of f , is shown.

a. Function f is not differentiable at two values of x on the interval $[-4, 4]$. Identify the two x -values and use the graph of f' to explain why f is not differentiable at these locations.

b. Write an inequality to compare the values $f'(-3)$, $f'(-1)$, $f'(3)$.

c. Find $f'(2)$.



9. If f is the function $f(x) = \ln(x^2 + 2) - \frac{1}{5}x^4 + 15x$, at which of the following values of x does f have the greatest instantaneous rate of change?
- A) $x = 0$
- B) $x = 1$
- C) $x = 2$
- D) $x = 3$
10. If $\lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} = 3$, then which of the following statements is NOT guaranteed?
- A) $\lim_{x \rightarrow 2} f(x) = f(2)$
- B) The slope of the line tangent to f at $x = 2$ is equal to 3.
- C) $\lim_{x \rightarrow 2^-} f'(x) = \lim_{x \rightarrow 2^+} f'(x)$
- D) All of the above statements are guaranteed.